Muhammad Nabeel Malik

Full-Stack AI Engineer | LLM Agents & Full Stack Development

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LinkedIn | GitHub | Portfolio

Professional Summary

Full-stack software engineer with 5+ years of experience building SaaS, fintech and e-commerce products, with a focus on real-time, data-heavy systems. Now working on AI engineering: LLM agents, RAG and tool calling in production. Main stack: Python, TypeScript, Nest.js, React.js, Next.js, PostgreSQL, Kafka, AWS and Docker.

Work Experience

Full-Stack AI Engineer (full-time), *OSPHIM* 07/2026 – Present
Aachen, Germany

- Migrated the AI agent from a self-hosted Qwen stack (LlamaIndex, ReAct) to the Anthropic Claude API, removing the legacy engine and redundant endpoints.
- Built a FastAPI backend and 12 native agent tools on Claude Sonnet 5.5 for Claire, letting operators log defects, write shift handovers and edit assets from chat, with human confirmation before every write.
- Designed the agent harness to report only what it can verify, e.g. "detection not configured" instead of a false "no anomaly", and kept user PII out of model context.
- Built evals that test Claire's answers against real production questions, catching regressions before each release.
- Built the shift-handover module end to end, from shift plans with full history and crew rostering to chat and voice entries, notifications and images, used across 3 plants and 40+ shifts per week.
- Shipped 56 merged PRs in a TypeScript monorepo built on NestJS, SvelteKit 5, PostgreSQL with Drizzle, and Redis, owning features from schema to UI and backing them with 250+ unit, integration and E2E tests in Vitest and Playwright.
- Made real-time machine-data views up to 160x faster by moving them from a 1.4-billion-row table to roll-up tables; 30-day dashboards load in 16 ms instead of 1,150 ms.

Full-Stack Engineer (part-time), *OSPHIM* 01/2026 – 07/2026
Aachen, Germany

- Built the first version of the AI assistant on self-hosted models: Qwen 3.5 35B for chat and handover extraction, Qwen 2.5 1.5B for cheap routing, and LlamaIndex ReAct agents to drive tools through prompting, since Qwen had no reliable native tool calling; answers drew on a Neo4j knowledge graph of problems, causes and countermeasures.
- Built voice handovers on self-hosted Whisper large-v3, moved from a local GPU to a remote model host, with name recognition and fuzzy matching to map spoken machine names to assets.
- Rebuilt the production logbook as a real-time feed with NestJS, WebSockets and Svelte, where one entry can link machines, molds, jobs, users and production cells.
- Migrated all logbook entries from plain text to JSONB using Drizzle ORM, with zero downtime and no data loss.
- Extended experiment setup from single-cavity to multi-cavity molds, so a 16-cavity mold needs one setup instead of 16.

Software Engineer, *MobileLive* 05/2024 – 10/2025
MonieCorp (Fintech Product) Remote

- Developed responsive and modular UI components in React + TypeScript + ShadCN, improving page load speed by 25% and reducing UI bugs by 40% after refactoring.
- Built 15+ SOLID-compliant React components for Admin panel using Cursor IDE for rapid development, cutting client feedback cycles from 3 days to 24 hours.
- Built multi-step React form for admins to configure tiered currency margins with validation; streamlined setup from 20+ minutes to <4 minutes per currency pair.

Staples Canada (MobileLive Client)

- Implemented Mixpanel tracking for 100,000+ users, capturing product views, add-to-cart, and checkout events, enabling business team to optimize flows and boost conversions by 12%.
- Built custom React features using Canva API for Staples Online Print Shop, enabling dynamic product personalization and increasing user engagement by 20%.
- Fixed 30+ production bugs within 4-hour SLAs, maintaining 99.9% uptime during peak sales.

Associate Software Engineer, *Code-Huddle* 08/2022 – 05/2024
GYMYG: Virtual Gym Partner (US-Based Startup) Islamabad, Pakistan
- Built Nest.js APIs with JWT RBAC, handling 10K DAU at <200ms P95 latency.
- Integrated Stripe monthly subscriptions, processing $15k+ in secure payments.
- Reduced video load times by approximately 30% using FFMPEG compression and optimized AWS CloudFront edge delivery.
- Automated end-to-end CI/CD pipeline with Github Actions, AWS CodePipeline, cutting deployment time by 40%.

Ordinami: QR Ordering Web-App (Italian Startup MVP)
- Implemented Firebase Auth + Firestore backend, supporting 100+ daily orders.
- Built an admin-panel shadcn/ui + Tailwind for menu creation, reducing manual setup time for owners by 60%.

Product Development Intern, *Code-Huddle* 04/2022 – 06/2022
Islamabad, Pakistan
- Assisted in frontend (React.js) and backend (NestJS) development, contributing to the design and implementation of scalable solutions.
- Supported AWS operations and collaborated on ongoing projects, gaining exposure to the full product development lifecycle.

Education

Bachelor of Science in Software Engineering, *COMSATS University* 2018 – 2022
Islamabad, Pakistan

Skills and Core Competencies

AI & LLM Engineering
Python | FastAPI | SQLAlchemy | Claude API | Qwen / Ollama | LlamaIndex | LLM Agents | Tool Calling | RAG | Evals | Whisper | Neo4j

Frontend Development
React.js | Next.js | SvelteKit | Context API | Tailwind CSS | HTML5 | CSS3 | Shopify (Liquid Templating) | ShadCN UI | Redux and Redux Toolkit | RxJs

Software Development & Engineering
Full-Stack Web Applications | RESTful APIs | Microservices Architecture | Secure Authentication Systems | Payment Integrations (Stripe) | CI/CD Automation | SaaS Product Engineering

Cloud, DevOps & Containerization
AWS (ECS, ECR, EC2, RDS, S3, CloudFront, CodePipeline, CloudWatch, SNS) | Azure | Docker

Backend Development
JavaScript | TypeScript | Node.js | Express.js | Nest.js | TypeORM | Prisma | Drizzle ORM | Kafka | Redis

Databases & Data Management
MySQL | PostgreSQL | MongoDB | Query Optimization | Data Modeling

Notable Freelance Projects

Pendfunds.com (M.V.P.), *Contract Management & Mediation Software* 01/2024 – 09/2024
- Developed full escrow payment flow with Stripe Connect and Optimus Bank merchant payouts in Nest.js, onboarding 100+ users and handling hundreds of transactions.
- Automated deployments across development, staging, and production environments using AWS Elastic Beanstalk and AWS CI/CD pipelines.
- Implemented secure payload encryption/decryption using PGP with RSA-256, ensuring sensitive transaction data remained fully protected.